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Cylinder Machining Instruction for Oversize Idler Shaft

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1.0 Products Affected

- QSK45 CM500
- QSK60 CM500
- QSK60 CM850 MCRS
- QSK60 DEP CM2150 MCRS
- QSK60 CM2150 MCRS
- QSK60 CM2250 K112
- QSK60 CM2350 K116
- QSK60 CM2350 K121
- QSK60 CM2350 K135
- QSK60 CM2350 K136
- QKS60 G CM2358 K110
- QSK78 CM500
- QSK78 CM2350 K154C
- QSK78 CM2250 K104

1.1 Overview

On some cylinder blocks the idler shaft bores were found damaged during service events. In order to be able to reuse the cylinder block, two kits of oversize idler shafts were created for QSK45/QSK60 and QSK78 (all except K126 and K137 series). There is also a QSK78 Accessory Comp oversize Idler Shaft for QSK78 CM2350 K154C engines which have the MCRS Fuel system

upgrade. This document provides the information on how to oversize the idler shaft bore and use oversize idler shaft kit. Reference Figure 1, Table 1, and Table 2 below for parts familiarization with the cylinder block and the current idler.

If the oversize of one of the idler shaft bores is required, the four idler shaft bores will be oversized (Dimension D on Figure 4) and the four current idler shafts and screws replaced by the oversize idler shaft kit.

For QSK78 CM2350 K154C engines which have the MCRS Fuel system upgrade, Idler Shaft, Part Number 6315345, and kit, Part Number 5628813, will need to be ordered. The accessory comp idler shaft, Part Number 5467076, of kit, Part Number 5628813, will be replaced by part number 6315345. Reference Figure 4 for existing and oversize shaft drawing specifications.

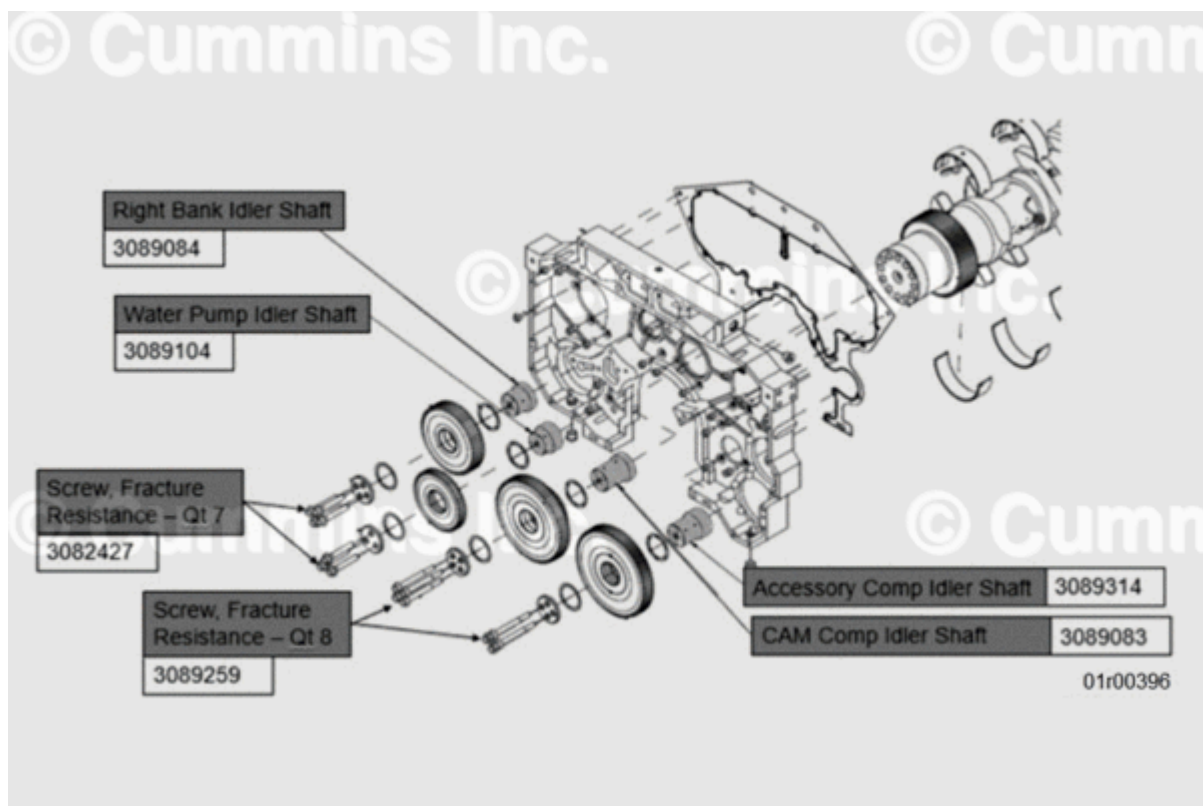


Figure 1, Parts Familiarization for QSK45/QSK60 Engine Models.

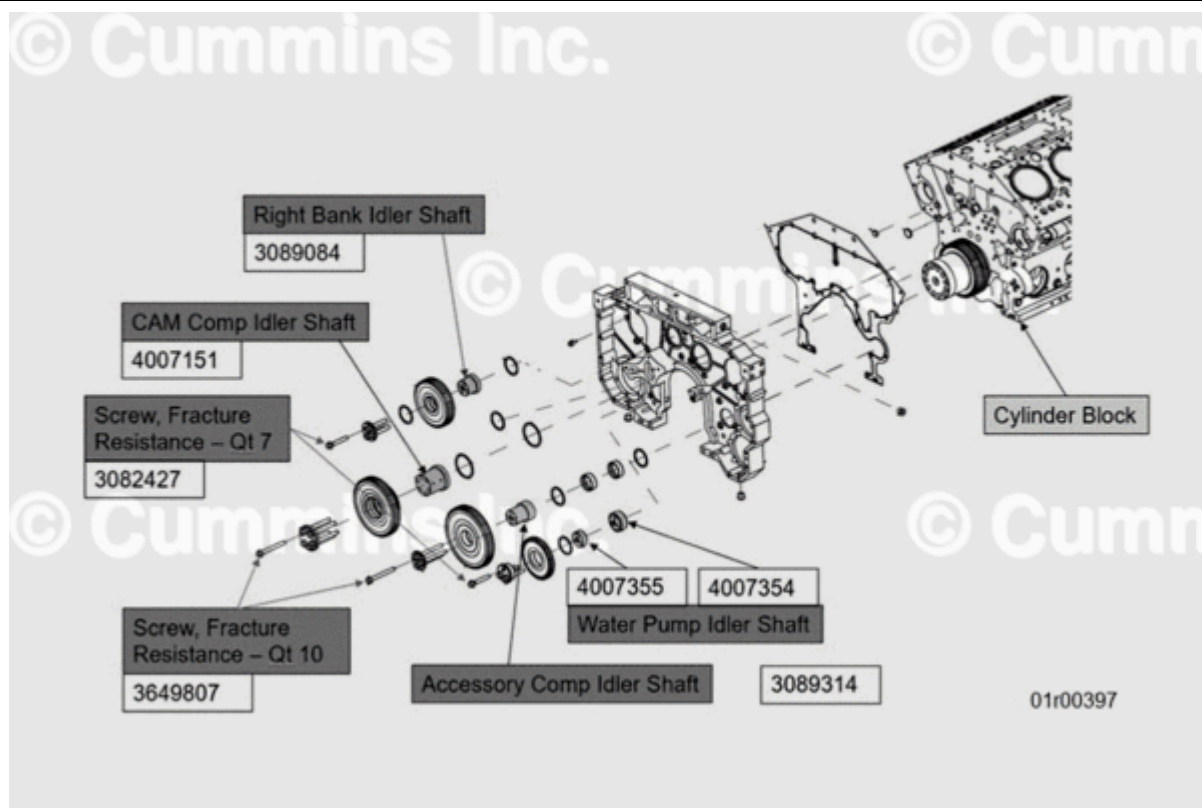


Figure 2, Parts Familiarization for QSK78 Engine Models.

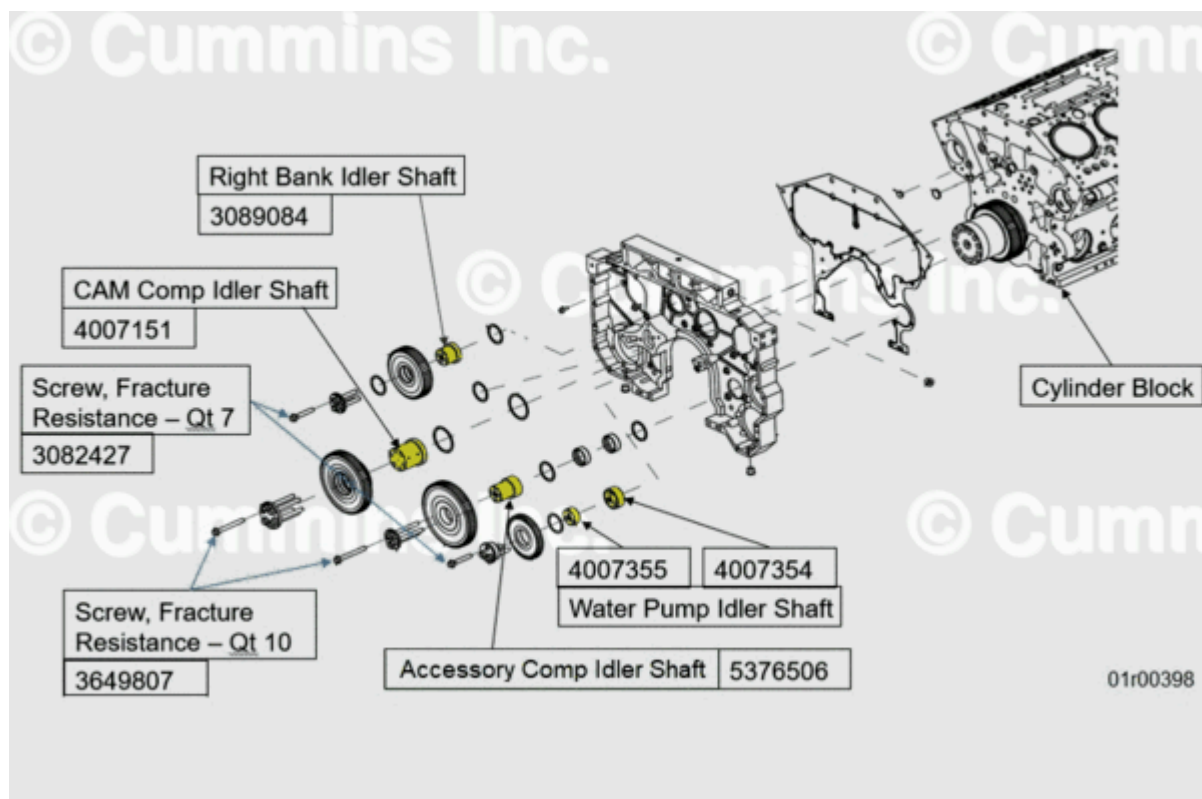


Figure 3, Accessory Comp Idler Shaft for QSK78 CM2350 K154C for Engines With the MCRS Fuel System Upgrade.

Table 1, QSK60 Service Kit for Oversize Idler Shaft				
Key	Part Description	Part Number (Oversize Shaft)	Part Number (Standard Shaft)	Quantity
1	Kit, Service	5581413	-	1
1.1	Idler,Shaft	5467064	3089083	1
1.2	Idler,Shaft	5467068	3089084	1
1.3	Idler,Shaft	5467072	3089104	1
1.4	Idler,Shaft	5467076	3089314	1
1.5	Screw,Fracture Resistance	3082427	3082427	7
1.6	Screw,Fracture Resistance	3089259	3089259	8

Table 2, QSK78 Service Kit for Oversize Idler Shaft				
Key	Part Description	Part Number (Oversize Shaft)	Part Number (Standard Shaft)	Quantity
1	Kit, Service	5628813	-	1
1.1	Idler,Shaft	5467060	4007354	1
1.2	Idler,Shaft	5467068	3089084	1
1.3	Idler,Shaft	5467076	3089314	1
1.4	Idler,Shaft	5467080	4007151	1
1.5	Idler,Shaft	4007355	4007355	1
1.6	Screw,Fracture Resistance	3082427	3082427	7
1.7	Screw,Fracture Resistance	3649807	3649807	10

Table 3, QSK78 CM2350 K154C Kit and Engines Accessory Comp Idler Shaft				
Key	Part Description	Part Number (Oversize Shaft)	Part Number (Standard Shaft)	Quantity
1	Idler,Shaft	6315345	5376506	1
2	Kit, Service*	5628813	-	1

*Note that when kit part number 5628813 is used for the HPI to MCERS upgrade, the idler shaft, Part Number 6315345, will replace part number 5467076. Therefore, idler shaft, Part Number 5467076, can be discarded.

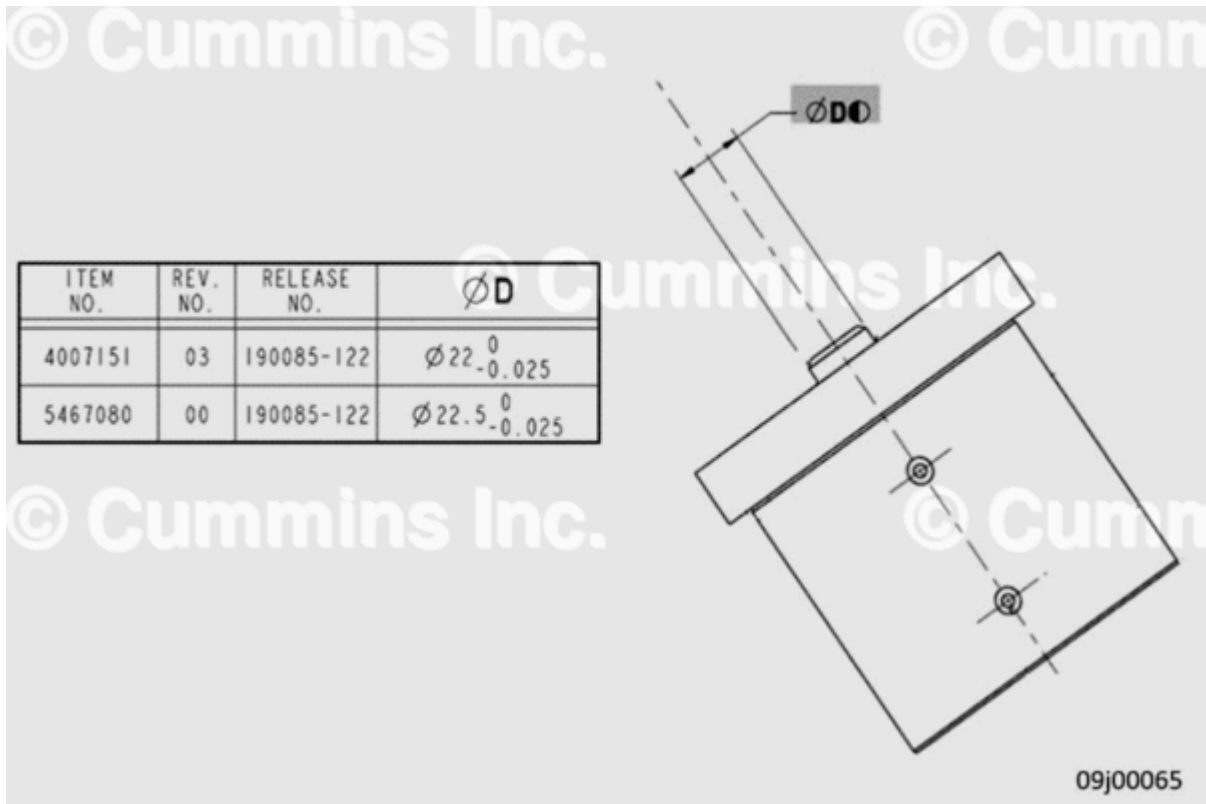


Figure 4, Example of Tab Drawing 5467058 with the Dimension D Standard and Oversized.

1.2 Procedure Selection

Use the service kit with oversize idler shaft (Part Number 5581413 for QSK60 and Part Number 5628813 for QSK78). All the idler shafts and screws **must** be replaced.

1.1.1 Validate Existing Bore Position

- Clean all idler bores thoroughly, remove debris, and degrease.
- Remove burrs and raised metal from idler bores around damage areas.
- Fit Pins Part A pin to each bore.
- Hammer into place with slide hammer to full depth of finished hole beyond damage.
- Select large digital Vernier scale and measure across pins to establish the radial distances.
- Calculate the actual distance by subtraction of the half pin dimension and record.
- Reference the data shown in Table 3 to confirm the bores are in position.
- If all dimensions are within specification, proceed to step 2. If errors found in measurements, reverify, and calculate adjustment needed to correct bore position before proceeding.

1.1.2 Components Set Up and Cut Oversize to Verified Positions

- Clean the cylinder block around the lubricating oil pan face, set onto machine and set up square to machine axis for front face.
- Verify by sweeping the front face of the cylinder block with an indicator clock, adjust until front face is square to axis.
- Move spindle location to in front of idler bores in turn with pins located.
- Use a centricator device in the machine spindle to adjust machine position and find the centre of the Part A pin diameter (see Part A drawing in appendix).
- Lock the machine axis position, move the spindle away laterally from the pin.
- Remove the centricator device from the spindle.
- Remove the Part A pin with a slide hammer.
- Keeping the machine position, fit an appropriate boring bar to spindle. Use machine to cut idler bore to oversize tolerance.
- Validate and adjust final bored size as necessary to achieve oversize.
- Repeat for each bore in turn.

1.1.3 Validate Final Bore Size and Position

- Once machined and final bore sizes verified and recorded, reconfirm achieved idler bore positions by use of the following steps.
- Repeat section one measurement checks process, use Part B pin (see Part B drawing in appendix) inserted into the oversize idler bores.
- Complete the records chart (see appendix 1.2.4) in full to confirm calculated positions are in expected tolerance.

1.1.4 Identification and Component Marking

- Oversize parts are different part numbers and are marked as oversize. Verify they are identified and kitted according to the correct cylinder block with corresponding bores.
- Mark the cylinder block as oversize "OS 0.5 mm" on the front face, adjacent to each idler location outside of the contact face, in readable 10 mm [3/8 in] stamped characters.
- Record in engine rebuild records, the use of the oversize set for future part ordering purposes.

1.1.5 Final Check to Confirm Achieved Gear Backlash Clearances are Achieved on Reassembly

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- See corresponding Service Manual. Reference Procedure 001-055 in Section 1, as a final check is performed.

1.3 Appendix

1.2.1 Gaging and Processing Equipment Requirements

- Suitable mill borer in good condition with axis locks, Centricator attachment with ± 0.002 mm increment scale.
- Fine boring bar with micron adjust to achieve final cut size.
- Comparator bore gage to measure bore size with 0.002 mm reading discernment.
- Large Vernier scale with digital reading, discernment ± 0.002 mm.
- Part A and Part B pins.
- Crank centre bar, ST number 3164872.
- Cam centre bar, 112 DIA g6 toleranced plain pin x 100mm long dimensions in the standard. (This cam centre bar is part of the kit ST-1177 / 3163649).

1.2.2 Service Tools Required

- Part A Pin x 4, Part B Pin x 4.
- Crank Centre Bar, Part Number 3164872.
- Cam Centre Bar Diameter 111.988mm MAX 111.966mm MIN, x 125 long.
- Boring machine in good condition with ideal horizontal spindle, or appropriate alignment vertical with 90-degree head attachment.
- Gages Required.
 - Large Vernier scale with digital reading, scale discernment ± 0.002 mm.
 - Digital Micrometer to check alignment pin calculation, scale discernment ± 0.002 mm.

1.1.1.1 Part A Drawing

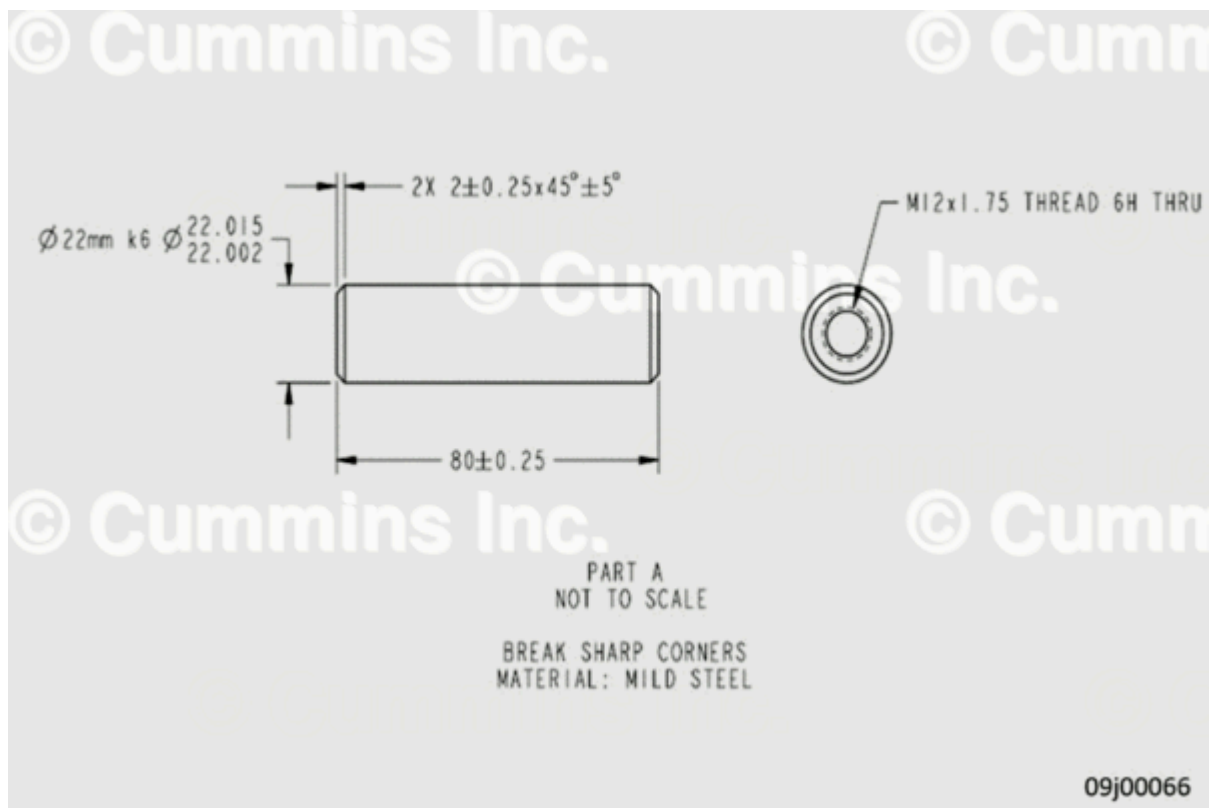


Figure 5, A Drawing of Part A.

1.1.1.2 Part B Drawing

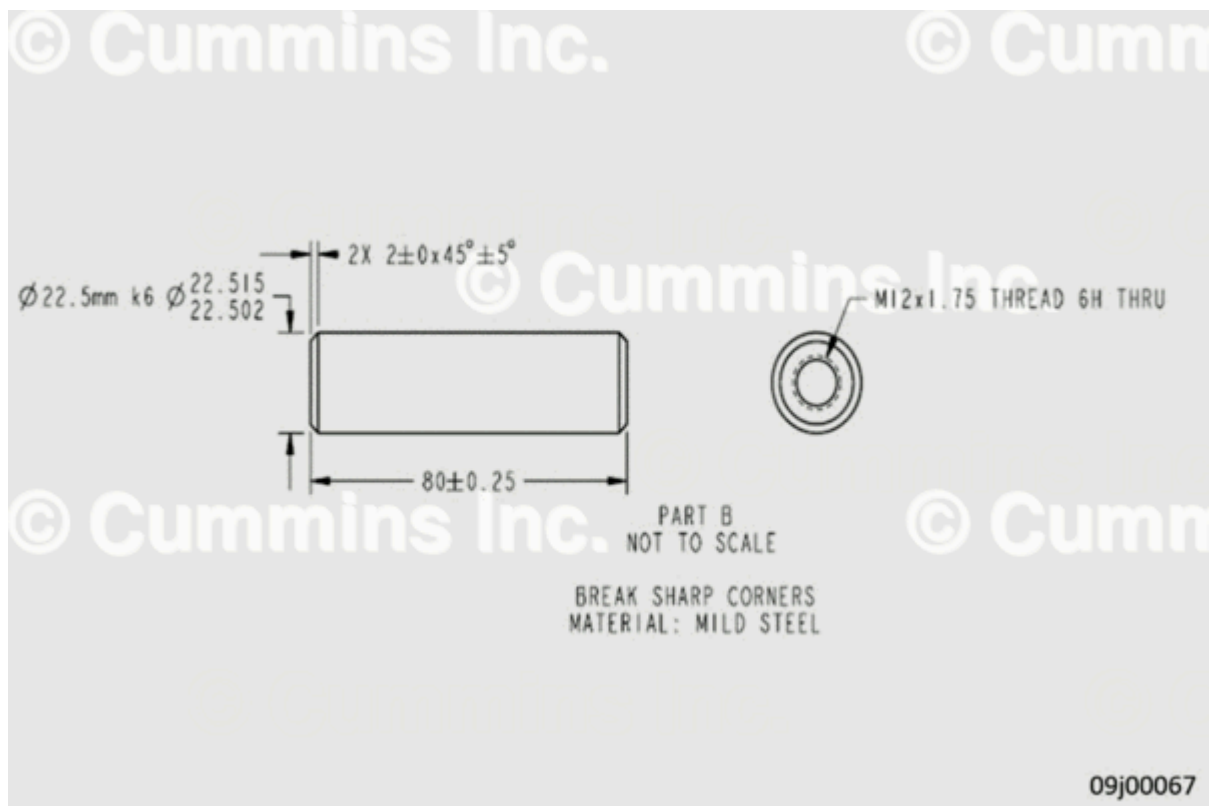


Figure 6, A Drawing of Part B.

1.1.1.3 Oversize Hole Size and Tolerance

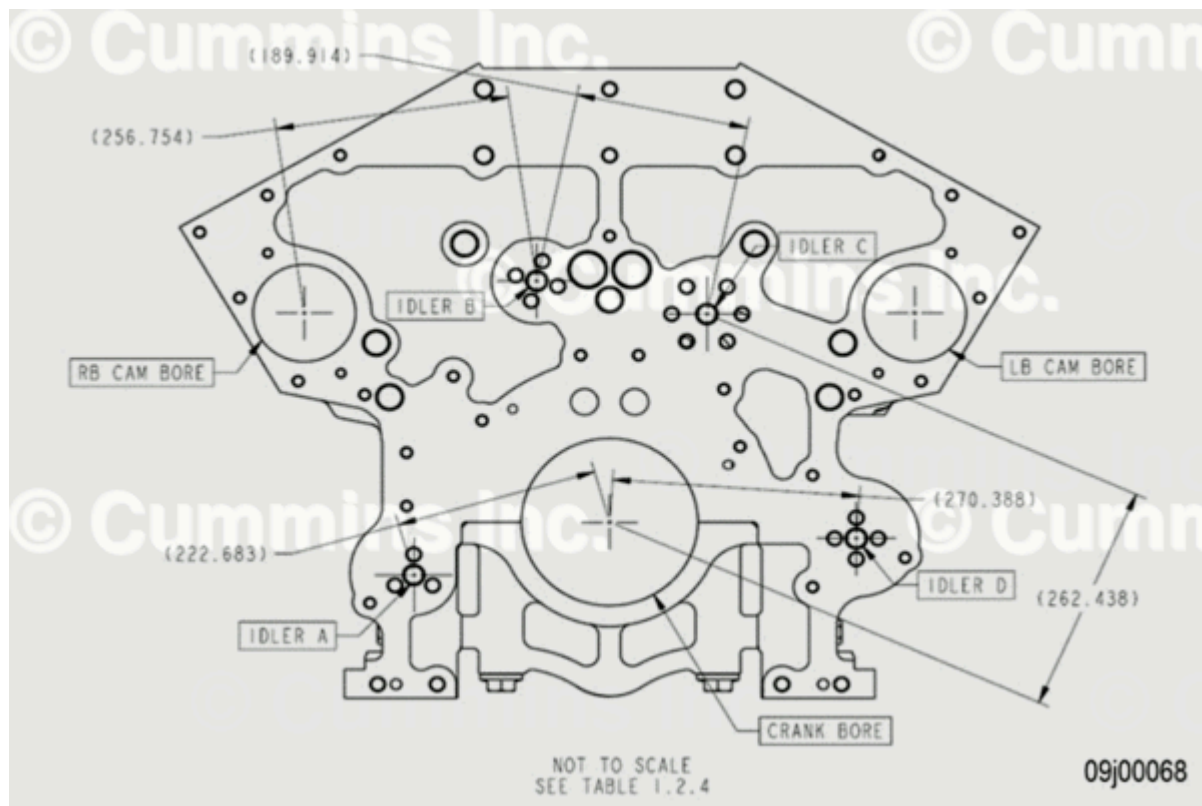


Figure 7, Idler Shaft Bore Internal Diameter Localization.

Table 4, Oversize Idler Shaft Bore Internal Diameter

	mm		in
Standard	21.975	MIN	0.865
0.5 mm [0.020 in]	22.475	MIN	0.885
	22.525	MAX	0.887
	22.025	MAX	0.867

Table 5, Measurement Validations

Idler Measurement	Radial Measurement	Off Machine Initial Measure (Part A Pin Fitted)	On Machine Post Cut Measure (Part B Pin Fitted)	Final Gear backlash as Measured on Assembly
Idler A to crank bore tolerance over pins	222.683 mm [8.76705 in] ± 0.035 mm [0.00138 in]	330.158 mm [12.99835 in] ± 0.035 mm [0.00138 in]	330.408 mm [13.00819 in] ± 0.035 mm [0.00138 in]	WP idler gear end clearance 0.330 mm [0.01299 in] to 0.150 mm [0.00591 in]

Table 5, Measurement Validations				
Record Idler A to crank bore actual				
Idler B to RB cam bore over pins	256.754 mm [10.1083 in] ± 0.035 mm [0.00138 in]	323.778 mm [12.74717 in] ± 0.035 mm [0.00138 in]	324.028 mm [12.75701 in] ± 0.035 mm [0.00138 in]	LB Camshaft to Idler back lash 0.070 mm [0.00276 in] to 0.051 mm [0.00201 in] MAX
Record Idler B to RB cam actual	-	-	-	-
Idler B to Idler C	189.914 mm [7.47693 in] ± 0.035 mm [0.00138 in]	211.949 mm [8.34445 in] ± 0.035 mm [0.00138 in]	212.449 mm [8.36412 in] ± 0.035 mm [0.00138 in]	Compound Gear to RB Idler back lash 0.070 mm [0.00276 in] to 0.051 mm [0.00201 in]
Record Idler B to RB cam actual	-	-	-	-
Idler C to Crank Bore	262.438 mm [10.33220 in] ± 0.035 mm [0.00138 in]	369.913 mm [14.56350 in] ± 0.035 mm [0.00138 in]	370.163 mm [14.57335 in] ± 0.035 mm [0.00138 in]	LB Camshaft to Idler back lash 0.070 mm [0.00276 in] to 0.051 mm [0.00201 in] MAX
Record Idler C to crank bore actual	-	-	-	-
Idler D to crank bore	270.388 mm [10.64520 in] ± 0.035 mm [0.00138 in]	376.863 mm [14.83713 in] ± 0.035 mm [0.00138 in]	377.113 mm [14.84697 in] ± 0.035 mm [0.00138 in]	-
Record Idler D to crank actual	-	-	-	-

Document History

Date	Details
2019-12-31	Module Created
2022-9-23	Updated Product Affected. Updated Overview, Figures 1, 2, and 3, and Tables 1, 2, and 3.

Last Modified: 30-Sep-2022
